

VARUN KUMAR K

M.Tech – Welding Engineering
Gender: Male
Date of Birth: 15/02/2001
E-mail : varunkumark68@gmail.com
Contact : 8088191103



Educational Qualification

Year	Degree/Examination	Institution/Board	CGPA/Percentage
2026	M.Tech Welding Engineering	National Institute of Technology, Tiruchirappalli	08.41*
2023	B. E Mechanical Engineering	JSS Science and Technology University, Mysore	07.77
2019	Class XII	Alvas Pu College, Moodbidri, Managalore	83.67%
2017	Class X	Shree Vidya Sagar High School, Yelahanka, KSEEB	88.48%

Internship Experience

- Internship at BHEL Trichy:** May 2025 – July 2025
Underwent a six-week internship at BHEL Trichy focused on welding processes and metallurgical evaluation. Investigated crack formation in SS347 overlay on 20MnMoNi55 using TIG welding. Additionally, observed advanced welding techniques such as 20-torch MIG/MAG, orbital welding and gained exposure to workshops specializing in boiler fabrication and critical components for thermal and nuclear power plants.

Projects

- Effect of Post-Processing on the Microstructural and Mechanical Properties of WAAM-Fabricated C250W Maraging Steel:** July 2025 – Dec 2025
My ongoing project focuses on optimizing heat treatment parameters (temperature and soaking time) to achieve chemical homogenization and uniform hardness distribution in WAAM-deposited Maraging Steel 250. The research involves post-processing through solution annealing and aging, followed by detailed characterization using SEM and TEM to analyze phase transformation, precipitate morphology, and microstructural evolution influencing the mechanical properties
- Analysing the Impact of Aging Treatment on Inconel 718 and changes in its Microstructure and Hardness:** Aug 2024 – Dec 2024
Conducted a detailed analysis of aging treatment effects on Inconel 718, focusing on microstructural transformations and hardness variations, studied the precipitation behaviour of strengthening phases like γ'' and δ phases during heat treatment, evaluated mechanical properties and correlated them with microstructural changes to optimize performance.

- Analysing the Effect of Solution Heat Treatment on the Microstructure of Duplex Stainless Steel 2205:** *Aug 2024 - Dec 2024*
 Air cooling after solution treatment at 1050 °C moderately preserves the δ -ferrite and γ -austenite balance in DSS 2205, maintaining its mechanical strength and corrosion resistance. However, it can lead to partial formation of intermetallic phases like sigma and chromium nitrides. These can reduce toughness, making phase control essential for reliable application. Air cooling remains practical but needs optimization.
- Solar Powered Sugarcane Harvesting Machine:** *Aug 2022 - July 2023*
 Designed and fabricated a solar-powered sugarcane harvesting machine operated by a rechargeable battery. The battery is charged via a solar panel, enabling clean and sustainable operation. The machine features two rotary cutters – one at the bottom for cutting at the root and one at the top for trimming cane height. It allows efficient, dual-point harvesting in a single pass. The system helps reduce manual effort and improve field productivity.

Areas of Interest

- Joining of Materials
- Welding Metallurgy
- Non-Destructive Testing
- Strength of Materials

Technical Skills and Certifications

Technical Skills:

- Engineering Software : Auto CAD
- Other Software : Microsoft Office

Certifications:

- Attended One-Day Workshop on “INTELLIGENT WELDING TECHNIQUES” organised by Indian Welding society, Southern zone at Tiruchirappalli.
- Geometric Dimensioning and Tolerancing (GD&T): Basics by Udemy

Positions of Responsibility

- Class Representative, Welding Engineering, NIT Trichy *Aug 2025 - Present*

Extracurricular Activities

- Successfully organized Pre Jayciana-Open Street Food Festival by coordinating with sponsors, volunteers. Managed event scheduling ensuring smooth operations with high attendance.
- I volunteered my college fest Jayciana by assuring with event coordination and management. I helped ensure smooth operations without any issues that arose. My contributions were crucial in making the fest a successful and enjoyable experience for all.